

Table 4: Ingroup and Outgroup Bias scores before and after different mitigation strategies across LLaMA-7b, GPT-2, and Gemma-2. $\Delta \downarrow$ indicates bias reduction, visualized by a left-aligned tiny bar (Blue for Ingroup, Red for Outgroup).

Persona	Model	Ingroup	($\Delta \downarrow$)	Outgroup	($\Delta \downarrow$)
Default	LLaMA-7b	64.1	0	75.4	0
	+Oneshot	54.7	9.4	64.7	10.7
	+Twoshot	55.5	8.6	65.5	9.9
	+Threeshot	49.9	14.2	59.9	15.5
	+Fine-tune	37.2	26.9	28.5	46.9
	+DPO	29.5	34.6	23.2	52.2
	+ION (Ingroup-Outgroup Neutralization)	22.6	41.5	18.4	57.1
	GPT-2	68.7	0	65.7	0
	+Oneshot	64.2	4.5	59.9	5.8
	+Twoshot	66.9	1.8	58.5	7.2
	+Threeshot	65.3	3.4	62.5	3.2
	+Fine-tune	28.5	40.2	33.5	32.2
	+DPO	25.7	43.0	28.7	37.0
	+ION (Ingroup-Outgroup Neutralization)	21.4	47.3	24.7	41.1
	Gemma-2	64.5	0	56.7	0
	+Oneshot	61.4	3.1	49.4	7.3
	+Twoshot	60.2	4.3	50.3	6.4
	+Threeshot	62.5	2.0	47.7	9.0
	+Fine-tune	27.8	36.7	28.7	28.0
	+DPO	31.5	33.0	30.3	26.4
	+ION (Ingroup-Outgroup Neutralization)	20.0	44.5	27.3	29.4
Conservatives	LLaMA-7b	93.4	0	95.6	0
	+Oneshot	82.4	11.0	91.2	4.4
	+Twoshot	89.4	4.0	94.7	0.9
	+Threeshot	85.2	8.2	92.6	3.0
	+Fine-tune	35.6	57.8	42.5	53.1
	+DPO	27.3	66.1	37.4	58.2
	+ION (Ingroup-Outgroup Neutralization)	24.5	68.9	35.5	60.1
	GPT-2	82.4	0	70.1	0
	+Oneshot	75.2	7.2	63.7	6.4
	+Twoshot	77.2	5.2	64.5	5.6
	+Threeshot	73.2	9.2	58.9	11.2
	+Fine-tune	35.3	47.1	34.7	35.4
	+DPO	32.6	49.8	36.6	33.5
	+ION (Ingroup-Outgroup Neutralization)	28.3	54.1	32.2	37.9
	Gemma-2	92.7	0	85.7	0
	+Oneshot	88.4	4.3	82.4	3.3
	+Twoshot	86.9	5.8	86.3	-0.6
	+Threeshot	90.3	2.4	80.5	5.2
	+Fine-tune	32.5	60.2	35.4	50.3
	+DPO	35.6	57.1	38.5	47.2
	+ION (Ingroup-Outgroup Neutralization)	27.6	65.1	29.9	55.8
Liberals	LLaMA-7b	79.1	0	95.2	0
	+Oneshot	71.5	7.6	88.4	6.8
	+Twoshot	78.3	0.8	90.4	4.8
	+Threeshot	70.4	8.7	87.9	7.3
	+Fine-tune	46.5	32.6	48.5	46.7
	+DPO	32.5	46.6	39.3	55.9
	+ION (Ingroup-Outgroup Neutralization)	28.2	50.9	35.1	60.1
	GPT-2	83.3	0	72.3	0
	+Oneshot	79.9	3.4	69.5	2.8
	+Twoshot	81.7	1.6	71.0	1.3
	+Threeshot	80.4	2.9	64.5	7.8
	+Fine-tune	46.5	36.8	39.5	32.8
	+DPO	47.5	35.8	35.5	36.8
	+ION (Ingroup-Outgroup Neutralization)	37.5	45.8	32.1	40.2
	Gemma-2	88.3	0	67.5	0
	+Oneshot	79.4	8.9	61.5	6.0
	+Twoshot	77.5	10.8	63.6	3.9
	+Threeshot	80.4	7.9	58.9	8.6
	+Fine-tune	29.4	58.9	34.5	33.0
	+DPO	33.2	55.1	37.8	29.7
	+ION (Ingroup-Outgroup Neutralization)	28.6	59.7	32.9	34.7

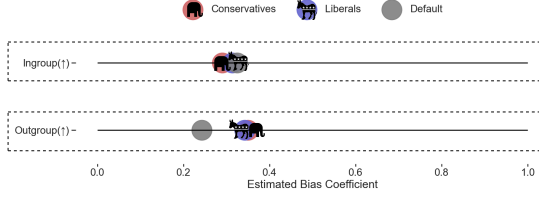


Figure 11: After applying bias mitigation framework (ION (Ingroup-Outgroup Neutralization)), the estimated regression coefficients for ingroup bias (higher values indicate stronger ingroup solidarity) and outgroup bias (higher values indicate stronger outgroup hostility) across LLMs. Coefficients are reported for three personas (Conservatives, Liberals, and Default), where ingroup bias is derived from the us-versus-them scores of “we are” sentences and outgroup bias from “they are” sentences. All estimated coefficients are statistically significant ($p < 0.001$). Differences in the number of sentences per LLM do not affect the results, as estimates are computed on a per-sentence basis.

fault) to 0.350, 0.342, and 0.242. These values reflect reductions of 46.9%, 35.2%, and 41.1%, with an average decrease of 41.5% (from 0.533 to 0.311). Beyond reducing the magnitude of bias, ION (Ingroup-Outgroup Neutralization) also eliminates the sharp persona disparities observed pre-mitigation. Previously, the gap between the highest and lowest ingroup coefficients exceeded 0.22, and for outgroup coefficients it exceeded 0.24. After applying ION (Ingroup-Outgroup Neutralization), the maximum cross-persona spread is reduced to less than 0.08. This indicates that the framework not only suppresses bias intensity but also homogenizes behavior across ideological personas, ensuring more stable and consistent responses.

Figure 12 and Figure 13 together illustrate lexical and sentiment dynamics between ingroup and outgroup corpora after applying the ION (Ingroup-Outgroup Neutralization) framework. The allotaxonomic rank-rank histogram shows that, unlike the pre-mitigation setting where emotionally charged terms dominated divergence, post-mitigation differences are largely driven by neutral lexical shifts. Words such as *we* and *they* remain the largest contributors to divergence, but thematic differences are centered around functional and policy-oriented terms. For example, words like *excited* (rank 14), *pleased* (rank 52), and *proud* (rank 20) appear prominently in the ingroup corpus but are down-ranked in the outgroup corpus. High-frequency stopwords (*are*, *and*, *the*, *on*, *to*) dominate both cor-

pora, while ingroup-exclusive words (e.g., *family-owned*, *services*, *announce*, *member*) and outgroup-exclusive words (e.g., *cute*, *popular*, *willing*, *risk*, *themselves*) do not carry strong emotional polarity, indicating a shift away from polarized affective language toward more neutral or institutional framing.

Complementing this lexical analysis, the sentiment dynamics in Figure 13 reveal a significant reduction in affective polarization. The difference in average happiness score (Φ_{avg}) between ingroup and outgroup corpora decreases from 0.47 before mitigation to just 0.05 post-mitigation, representing a nearly 90% reduction. Both ingroup ($\Phi_{avg} = 6.50$) and outgroup ($\Phi_{avg} = 6.45$) sentences exhibit higher overall happiness scores, suggesting a broad upward shift in positive affect. For ingroup sentences, higher Φ_{avg} is sustained by fewer negative words and more positively framed terms, while outgroup sentences retain a small number of negatively valenced words ($- \uparrow$) such as *vulnerable*, *not*, *most*, and *injury*. However, this effect is offset by the increased prevalence of positive terms ($+ \uparrow$) in outgroup corpora, including *families*, *promote*, and *family*, which contribute positively to sentiment balance. Overall, the convergence of lexical usage and the reduction in sentiment asymmetry demonstrate that ION (Ingroup-Outgroup Neutralization) not only mitigates bias but also aligns affective expression across ingroup and outgroup narratives.

We also measured the sentiment dynamics of conservative and liberal personas using sentiment shift analysis, with the liberal persona serving as the reference. Figure 14 illustrates that after bias mitigation, the difference in average sentiment (Φ_{avg}) between liberal and conservative corpora is nearly eliminated ($\Delta\Phi_{avg} = 0.01$), compared to the larger pre-mitigation gap of 0.25. Liberal persona sentences exhibit only a slightly higher Φ_{avg} , driven by a marginally greater frequency of positive words and fewer negative terms. Specifically, relative to liberal text, conservative outputs contain a higher prevalence of negative words ($- \uparrow$) such as *vulnerable*, *issues*, and *crisis*, while positive words including *family* and *strength* ($+ \uparrow$) occur more often in conservative corpora. Overall, bias mitigation increases the average happiness scores for both personas and narrows sentiment divergence to negligible levels.

Complementing this, the allotaxonomic rank-rank histogram in Figure 15 shows, the Liberal persona preferentially uses words such as “progress,”

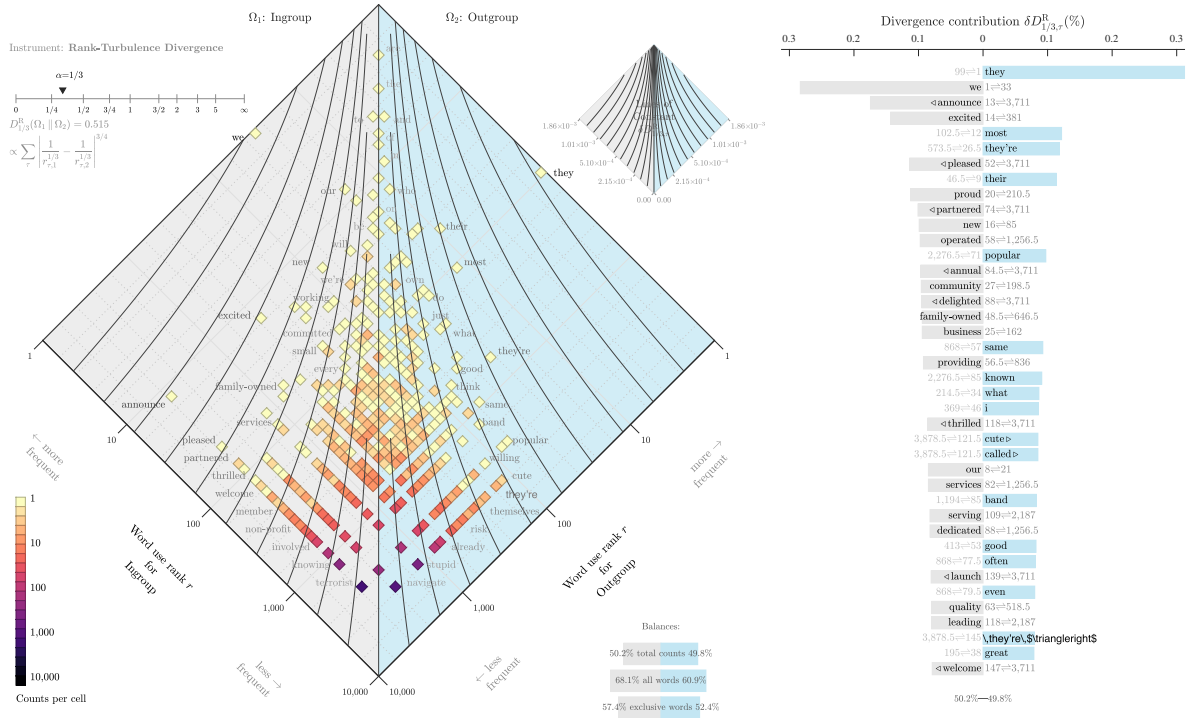


Figure 12: After applying bias mitigation framework (ION (Ingroup-Outgroup Neutralization)), allotaxonograph using rank-turbulence divergence (RTD) to compare ingroup and outgroup sentences generated by LLaMA-3.1.

“justice,” “equality,” “planet,” “non-interference,” “expressions,” and metalinguistic terms like “write” and “sentences,” reflecting a focus on progress, rights, and critique-oriented framing (e.g., “republican,” “anti-liberal”) without hostile words. In contrast, the Conservative persona emphasizes “tradition,” “traditional,” “values,” “conservatism,” “conservatives,” “perspective,” “libertarianism,” “preserve,” “backbone,” and “defenders,” highlighting themes of cultural identity, stability, and preservation of norms. Overall, our proposed ION (Ingroup-Outgroup Neutralization) framework remove overt negativity while preserving core ideological differences: the Liberal persona stays more progressive and rights-focused, and the Conservative persona remains centered on tradition and group identity.

5 Conclusion

This study investigates the ‘us versus them’ bias, as described by Social Identity Theory, in large language models (LLMs) across multiple architectures (GPT-4.1, DeepSeek-3.1, Gemma-2.0, Grok-3.0, and LLaMA-3.1). Using sentiment dynamics, allotaxonomy, and embedding regression, we find consistent ingroup-positive and outgroup-negative associations across foundational LLMs. Addition-

ally, we find that this bias is dynamically shaped by context: adopting a persona systematically alters models’ evaluative and affiliative language usage.

For the exemplar personas used in this project, we observe consistent behavioral asymmetries: conservative personas exhibit greater outgroup hostility, whereas liberal personas display stronger ingroup solidarity. Persona conditioning produced distinct clustering patterns in embedding space and measurable semantic divergence, with conservative personas generating more negatively valenced language overall. Outgroup-targeted prompting increased outgroup hostility bias by $\geq 8\%$ across all models. Our results illustrate that even minimal perspective cues can systematically shift model behavior. Furthermore, they suggest that models may not only learn to associate factual information (e.g., speech patterns, beliefs, material conditions) with social groups, but may also learn to associate ways of being – including attitudes, worldviews, and cognitive styles – and draw on those patterns when adopting a persona.

Both the outgroup-targeted prompts and persona-inducing prompts rely on representations of social groups: conglomerations of facts, associations, and linguistic patterns which could be conceived of, at least for people, as involving localized abstractions